IBM FileNet P8 V5.5.8

Detailed Study Material

Day 6 – Process Engine & Workflow Architecture

PE Overview | Key Components | Workflow Concepts | Installation & Configuration

Author	Course Day	Version
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Section 1 | Learning Objectives

By the end of Day 6, students will be able to:

1	Describe the IBM FileNet Process Engine V5.5.8 architecture and its role in the FileNet P8 platform
2	Identify and explain all key Process Engine components: Workflow System, Work Objects, Event Actions, and Queues
3	Explain workflow automation capabilities including event-driven launch, routing rules, and step processors
4	Define Workflow Concepts: work items, step processors, rosters, isolated regions, and workflow routing
5	Install IBM FileNet Process Engine V5.5.8 on Windows Server with WebSphere Application Server
6	Create and configure the PEDB database in DB2 for Process Engine
7	Configure a PE Connection Point linking the Content Engine domain to the Process Engine
8	Create workflow queues (rosters and work queues) in the ACCE Process Configuration Console
9	Verify Process Engine installation using ACCE and the PE Administration Tool
10	Troubleshoot common Process Engine installation and connectivity issues

Section 2 | Process Engine Overview

2 Process Engine Overview

IBM FileNet Process Engine (PE) V5.5.8 is the workflow automation server in the FileNet P8 platform. It manages the lifecycle of business processes by routing work items through defined workflow steps, assigning tasks to users or groups, enforcing business rules, and integrating with external applications.

Process Engine in the FileNet P8 Architecture

- PE is a separate server component deployed as a Windows service (IBM FileNet Process Engine Service).
- PE communicates with the Content Engine (CE) via RMI/IIOP on port 32768.
- PE stores its data in a dedicated database: PEDB (DB2 or Oracle).
- PE workflow definitions are created in IBM Process Designer and deployed to the PE server.
- End users interact with workflows through IBM Content Navigator (ICN) or the legacy Workplace XT.
- Administrators manage PE via ACCE > Process Configuration Console or the PE Admin Tool.
- In our training environment: PE V5.5.8 runs on the same Windows server as CE and WAS.

2.1 Workflow Automation Capabilities

The Process Engine provides a comprehensive set of workflow automation features:

Capability	Description
Event-Driven Launch	Workflows can be automatically launched when FileNet document events fire (OnCreation, OnCheckin, OnDelete) via Event Subscriptions with Process Router action.
Manual Launch	Users can manually launch workflow instances from ICN, the CE Java API, or the PE REST API.
Routing Rules	Conditional routing directs work items to different queues or users based on document properties, user input, or custom expressions.
Parallel Processing	Multiple workflow branches can execute simultaneously using parallel split/join steps.
Deadline Management	Each workflow step can have configurable deadlines. Overdue items are escalated automatically.
Sub-Workflows	A parent workflow can launch child workflows and wait for their completion before continuing.
External Integration	PE can call external web services, Java classes, or execute system commands via Custom Step Processors.
Audit Trail	Every workflow action (launch, step, reassign, complete) is logged in the PEDB with timestamp and user.
Work Item Reassignment	Supervisors can reassign overdue or blocked work items to other users or groups.
Milestone Tracking	Workflow milestones mark progress through defined business stages for reporting.

Section 3 | Key Components of Process Engine

3 Key Process Engine Components

3.1 Workflow System

The Workflow System is the top-level container in the Process Engine. It represents the entire PE server instance and contains all workflow definitions, rosters, queues, and configuration.

Component	Description
PE Server (Windows Service)	The IBM FileNet Process Engine Service running on Windows. Manages all workflow processing, queue management, and event routing.
Process Engine Database (PEDB)	DB2 database storing all workflow instance data, work item states, audit logs, roster definitions, and workflow map definitions.
Process Configuration Console	Web-based admin UI accessible via ACCE > Process Configuration Console. Used to create queues, rosters, connection points, and monitor workflows.
PE Administration Tool	Standalone Java tool for advanced PE management: server statistics, queue depth monitoring, isolated region management.
Workflow Router	Internal PE component that evaluates routing expressions and directs work items to the correct next step or queue.
Dispatch Engine	Multi-threaded PE component processing step transitions. Default thread pool: 10. Configurable for high-volume deployments.

3.2 Work Objects

A Work Object represents a running instance of a business process. When a workflow is launched, a new Work Object is created in the PEDB. Each Work Object has a unique Work Object Number.

3.2.1 Work Object Properties

Property	Description
Work Object Number	Unique numeric identifier assigned when the workflow instance is launched. Primary key in PEDB.
F_Subject	The primary subject/title of the workflow instance. Displayed in the work queue. Required for all workflows.
F_BoundDocument	Optional: a FileNet document ID attached to the workflow. Links the workflow to a specific document.
F_Deadline	Optional deadline date/time. PE evaluates this automatically and escalates overdue instances.

F_Originator	The username of the person or process that launched the workflow.
F_StepName	The current step name where the work item is waiting. Updated each time the item moves to a new step.
Custom Data Fields	Developer-defined fields in the workflow map (e.g., ApprovalStatus, DocumentType, Priority). Stored in PEDB.
Milestone	A named point in the workflow used for progress tracking. Set programmatically or via workflow steps.

3.2.2 Work Object Lifecycle

A Work Object progresses through a defined lifecycle from launch to completion:

Work Object Lifecycle States

1. LAUNCHED – Workflow instance created. F_Subject, data fields, and optional document attachment set.
2. IN_QUEUE – Work item is waiting in a queue for a user to claim it.
3. IN_PROCESS – A user has locked (claimed) the work item and is actively working on it.
4. STEPPING – PE is processing the step transition: evaluating routing expressions and moving to next step.
5. COMPLETED – All steps executed. Work Object moved to history/archive table in PEDB.
6. TERMINATED – Workflow terminated early by an administrator or an Abort step.
7. EXCEPTION – An error occurred in a system step (e.g., web service call failed). Requires admin intervention.

3.3 Event Actions

Event Actions are PE-specific actions that can be triggered by FileNet Content Engine events (via Event Subscriptions) or directly from within a workflow map.

3.3.1 Types of PE Event Actions

Event Action Type	Description	Notes
Process Router	Launched from a CE Event Subscription. When a document event fires (e.g., OnCheckin), PE automatically launches a configured workflow.	Most common. Used for document-driven automation.
Mailer	Sends an email notification when a workflow reaches a specific step. Configured with SMTP server settings.	Used for notifications and escalations.
Component Step	Calls a custom Java class (Component Queue) implementing the IComponentStep interface. For system integration.	Advanced: requires Java development.
Script Step	Executes a JavaScript expression within the workflow map. Can manipulate data fields and set routing variables.	Available in PE V5.5.8. No external compilation needed.

Abort Step	Immediately terminates the current workflow instance with status TERMINATED.	Used for error handling and early exit scenarios.

3.3.2 Process Router Configuration

The Process Router is the most important Event Action. It bridges the Content Engine event system with the Process Engine workflow system.

```
Process Router Configuration in ACCE:

1. Navigate to: TrainingOS > Events, Actions, Processes > Event Actions
2. Click New Event Action
3. Set Action Type = Process Router
4. Set Connection Point = PE_ConnectionPoint
5. Set Workflow Definition = TrainingWorkflow (the .pep file deployed to PE)
6. Set Launch Step = LaunchStep (the entry step in the workflow)
7. Map Data Fields:
   - F_Subject  <-- EventObject.DocumentTitle   (document title becomes workflow
subject)
   - F_BoundDocument <-- EventObject.Id          (links workflow to the document)
8. Click Save

9. Create an Event Subscription:
   - Navigate to: Events, Actions, Processes > Subscriptions > New Subscription
   - Event Class = OnCheckin, Object Class = Document
   - Timing = Post-processing (asynchronous)
   - Event Action = [the Process Router created above]
   - Enable the subscription
```

3.4 Queues (Rosters and Work Queues)

Queues are the holding areas where work items wait for user action. The Process Engine uses a two-level queue model: Rosters contain Work Queues.

3.4.1 Roster

A Roster is a top-level container that groups related work queues on a Process Engine server. Every PE server has a default roster (DefaultRoster). Multiple rosters can be created to separate different business units or workflow categories.

Property	Notes
Roster Name	Unique identifier on the PE server. Example: TrainingRoster. Alphanumeric, no spaces.
Description	Human-readable label shown in admin tools.
PE Server	The PE server instance this roster belongs to. In training: localhost:32768
Isolated Region	The isolated region this roster's queues belong to. Default: Region 1. One roster = one isolated region.
Capacity	Maximum number of concurrent workflow instances in this roster. Default: unlimited.

3.4.2 Work Queue

A Work Queue (also called a Step Queue) is where individual work items land when a workflow step requires human action. Users browse, claim, and step work items from their assigned queues in ICN.

Property	Notes
Queue Name	Unique within the roster. Example: ApprovalQueue, ReviewQueue. Displayed in ICN.
Queue Type	Classic (default) – FIFO ordering. Custom – application-controlled ordering.
Participant	The user or group allowed to work items in this queue. Mapped to LDAP group or individual user.
Deadline Expression	Optional: a PE expression that sets F_Deadline automatically when a work item enters this queue.
Escalation Rule	Automatic reassignment rule when deadline expires. Can reassign to supervisor or different queue.
Max Depth	Alert threshold. When queue depth exceeds this, PE can send notifications.

3.4.3 Queue Architecture Diagram

Process Engine Queue Hierarchy

```
PE Server (localhost:32768)
  |
  +-- DefaultRoster (Region 1)
  |     |
  |     +-- DefaultQueue         (generic catch-all queue)
  |     +-- ApprovalQueue        (documents awaiting manager approval)
  |     +-- ReviewQueue          (documents awaiting reviewer action)
  |
  +-- TrainingRoster (Region 1)
        |
        +-- TrainingApprovalQueue (created in Day 6 Lab)
        +-- TrainingReviewQueue   (created in Day 6 Lab)
```

Section 4 | Workflow Concepts

4 Workflow Concepts

4.1 Work Items

A Work Item is a single unit of work in a queue — one step of a workflow instance waiting for user action. When a workflow instance (Work Object) is at a step that requires human action, a Work Item is created in the target queue.

Work Item vs Work Object

Work OBJECT: The overall workflow instance. Has a unique Work Object Number. Persists for the entire workflow life.

Work ITEM: A specific step of the work object sitting in a queue. Created when a routing step reaches a queue step.

One Work Object generates one Work Item per queue step. Multiple parallel branches = multiple Work Items.

Work Item Property	Description
Work Item ID	Unique identifier for this step instance. Combination of Work Object Number + Step Number.
Queue Name	The queue this work item currently resides in. Changes when item is stepped to a different queue.
Locked By	Username of the user who has claimed (locked) this work item. Empty if available for claiming.
Date Received	When the work item arrived in the current queue.
Deadline	Optional deadline for this specific step. Can differ from the overall work object deadline.
F_Subject	Inherited from the Work Object. The primary description displayed in queue lists.
Step Data	All custom data fields defined in the workflow map (e.g., ApprovalStatus=Pending).

4.2 Step Processors

A Step Processor is the code that executes when a workflow step is processed. Different step types use different step processors.

4.2.1 Built-in Step Processor Types

Step Type	Description	Use Case
Queue Step	Places work item in a specified queue for human action. Most common step type.	ApprovalQueue, ReviewQueue

System Step	Executes automatically without human interaction. Runs PE expressions, scripts, or component steps.	Data field evaluation, routing logic
Component Step	Calls a custom Java class implementing IComponentStep. For external system integration.	ERP updates, email systems
Map Step	Launches a sub-workflow (child process) and optionally waits for it to complete.	Sub-process delegation
Notify Step	Sends an email notification using PE Mailer configuration. No queue assignment.	Approval notifications
Abort Step	Terminates the workflow instance immediately with status TERMINATED.	Error handling, rejection flows
Milestone Step	Sets a named milestone marker on the workflow instance for progress tracking.	Process tracking, SLA monitoring
Script Step	Executes a JavaScript expression to evaluate conditions and set routing variables.	Complex routing logic

4.2.2 Component Step Development (IComponentStep)

Custom Component Steps allow PE to integrate with external systems. The step is a Java class deployed as a JAR to the PE server.

```
// Minimal IComponentStep implementation
import filenet.vw.api.*;

public class MyComponentStep implements VWStepProcessor {
    public void doStep(VWStepElement step) throws VWException {
        // Get a data field value from the workflow
        String docTitle = (String) step.getFieldValue("F_Subject");

        // Perform business logic (e.g., call external API)
        boolean approved = callApprovalSystem(docTitle);

        // Set a result field to control routing
        step.setFieldValue("ApprovalResult", approved ? "Approved" : "Rejected");

        // Commit the step — moves to next routing decision
        step.commitStep(null);
    }
}
```

4.3 Workflow Routing

Workflow Routing controls how work items move from one step to the next. The Process Engine evaluates routing expressions at each step transition.

4.3.1 Routing Types

Routing Type	Description
Sequential Routing	Work item moves from one step to the next in a fixed sequence. No branching. Simplest pattern.
Conditional Routing	Route to different steps based on a Boolean expression or data field value. Example: If ApprovalStatus = Approved, go to PublishStep; else go to RejectionStep.
Parallel Routing (Split/Join)	Work item splits into multiple parallel branches, each processed independently. A Join step waits until all branches complete before continuing.
Iterative Routing (Loop)	Route back to a previous step when certain conditions are met. Example: Send document back to author if rejected (loop up to 3 times).
Milestone-Based Routing	Route based on milestone state. Used in complex multi-stage approval processes.

4.3.2 PE Routing Expressions

Process Engine uses a proprietary expression language for routing conditions. Expressions evaluate to TRUE or FALSE to determine the routing path.

```
// PE Routing Expression Examples

// Simple field comparison
F_ApprovalStatus == "Approved"

// Numeric comparison
F_DocumentCount > 5

// String functions
Strings.upper(F_Department) == "FINANCE"

// Date comparison
Dates.diff(F_Deadline, Dates.currentDate()) < 0   // Overdue check

// Logical operators
F_Priority == "High" && F_ApprovalStatus == "Pending"

// NULL check
F_ApprovedBy != null && F_ApprovedBy != ""
```

4.3.3 Workflow Map Structure

A Workflow Map is the visual design of a business process created in IBM Process Designer. It defines all steps, routing rules, data fields, and queue assignments.

Map Element	Description
Launch Step	Entry point of the workflow. Defines required launch parameters (F_Subject, data fields). Must have exactly one launch step.
Routing Decision	Diamond-shaped branch point. Evaluates a Boolean expression to choose the next step.

Queue Step	Rectangle with queue icon. Assigns work item to a queue for human action.
System Step	Rectangle with gear icon. Auto-executes without human interaction. Used for scripts and component steps.
Connector	Arrow linking steps. Can carry routing expressions for conditional routing.
Terminate	End point. When reached, the workflow instance is marked COMPLETED.

Section 5 | Process Engine Installation Overview

5 Process Engine Installation Overview

Before the hands-on lab, this section provides a conceptual overview of the PE installation steps. The detailed step-by-step instructions are in the Lab Guide.

5.1 Installation Prerequisites

Process Engine V5.5.8 Installation Prerequisites

Hardware: Minimum 4 GB RAM (8 GB recommended), 20 GB free disk space for PE installation
OS: Windows Server 2016 or 2019 (64-bit) — same server as CE in training lab
WAS: WebSphere Application Server 8.5.5+ running with FileNetServer profile (from Day 2)
CE: Content Platform Engine V5.5.8 installed and running (from Day 2)
P8Domain: FileNet P8 domain P8Domain must exist and be accessible
DB2: DB2 V11.5 running with GCDDB and OSDB already created (from Day 2-3)
PEDB: A new DB2 database for Process Engine will be created in the lab
Admin rights: Installation must run as local Administrator
IBM Installation Manager: IBM IM V1.9+ must be installed (used in Day 2)

5.2 Installation Architecture

Item	Details
PE Software Package	IFCPE_5.5.8_WIN_ML.zip (IBM FileNet Content Platform Engine Process Engine)
Install Directory	C:\IBM\FileNet\ProcessEngine\ (default)
PE Windows Service	IBM FileNet Process Engine (vwtaskd.exe) — starts automatically after installation
PE Database	PEDB — created in DB2 before running the PE installer
JDBC Data Source	ECMProcessDS — created in WAS Admin Console for PE to connect to PEDB
Connection Point	PE_ConnectionPoint — created in ACCE linking P8Domain to PE server
RMI/IIOP Port	32768 — default PE communication port. Must be open in Windows Firewall.
PE Admin Tool	vwadmin.exe — PE Administration Tool for monitoring and management

5.3 Post-Installation Configuration

After installing PE, the following configuration steps are required before workflows can be created and executed:

1. Create PEDB database in DB2 (pre-installation, covered in lab)
2. Create JDBC Data Source (ECMProcessDS) in WAS Admin Console pointing to PEDB
3. Run PE bootstrap utility to initialize the PEDB schema

4. Configure PE server settings (PE Admin Tool): RMI port, heap size, thread pool
5. Create Connection Point in ACCE: P8Domain > Process Engine > Connection Points
6. Create Isolated Region in ACCE (optional — default Region 1 is usually sufficient)
7. Create Roster and Work Queues in ACCE Process Configuration Console
8. Verify PE connectivity using ACCE > Process Configuration Console

Section 6 | Troubleshooting & Quick Reference

6 Troubleshooting & Quick Reference

6.1 Common PE Issues

Symptom	Likely Cause	Resolution
PE Service won't start	PEDB not created or JDBC data source misconfigured	Create PEDB in DB2. Verify ECMProcessDS in WAS Admin Console.
Connection Point shows Offline	PE service stopped or RMI port 32768 blocked	Start PE service (services.msc). Check Windows Firewall rule for port 32768.
Workflow won't launch from event subscription	Process Router event action misconfigured or PE connection point not selected	Verify event action configuration. Check PE connection point name matches.
Work item stuck in queue	User does not have queue access or PE dispatch thread pool exhausted	Add user/group to queue participants. Increase dispatch threads in PE config.
PE bootstrap fails	Wrong PEDB credentials or DB2 not accessible	Verify ECMProcessDS JNDI name matches PE config. Test DB connection in WAS.
Queue depth growing uncontrollably	Insufficient PE dispatch threads or component step errors	Check PE logs for step errors. Increase dispatch thread pool to 20-40.
VWException in application	Workflow data field mismatch or wrong field name	Verify field names in workflow map match the API call. Check PEDB schema.

6.2 Key Log Files

Log File	Contents
PE Server Log	C:\IBM\FileNet\ProcessEngine\log\vwtaskd.log — PE service startup, step processing, routing errors
PE Admin Tool Log	C:\IBM\FileNet\ProcessEngine\log\vwadmin.log — admin tool connection and operation logs
WAS SystemOut.log	C:\IBM\WAS\profiles\FileNetServer\logs\server1\SystemOut.log — JDBC errors, PE bootstrap output
DB2 Diagnostic Log	C:\IBM\DB2\DB2\logs\db2diag.log — DB2 connection and query errors for PEDB

6.3 Quick Reference

Item	Value	Notes

PE RMI Port	32768	Default. Configurable in PE Admin Tool.
PE Windows Service	IBM FileNet Process Engine	Start/stop via services.msc
PE Install Directory	C:\IBM\FileNet\ProcessEngine\	Default installation path
PEDB JNDI	ECMProcessDS	Must match PE bootstrap config
PE Admin Tool	vwadmin.exe	C:\IBM\FileNet\ProcessEngine\bin\
CE to PE Protocol	RMI/IIOP	Requires network access on port 32768
PE Bootstrap Command	vwbootstrap.exe	Run after PEDB creation to initialize schema
Connection Point Location	ACCE > P8Domain > Process Engine > Connection Points	Created after PE installation
ACCE Process Config	ACCE > Tools > Process Configuration Console	Queue/roster management
WAS JDBC Path	WAS Admin Console > Resources > JDBC > Data Sources	ECMProcessDS configuration